

Computer Graphics – October 29th, 2020

Name: _____

Seat Number: _____

Matriculation Number: _____

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (11 pages on 6 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If needed, we staple additional blank sheets.
- Only writing material is allowed!

I have read and understood the instructions above.

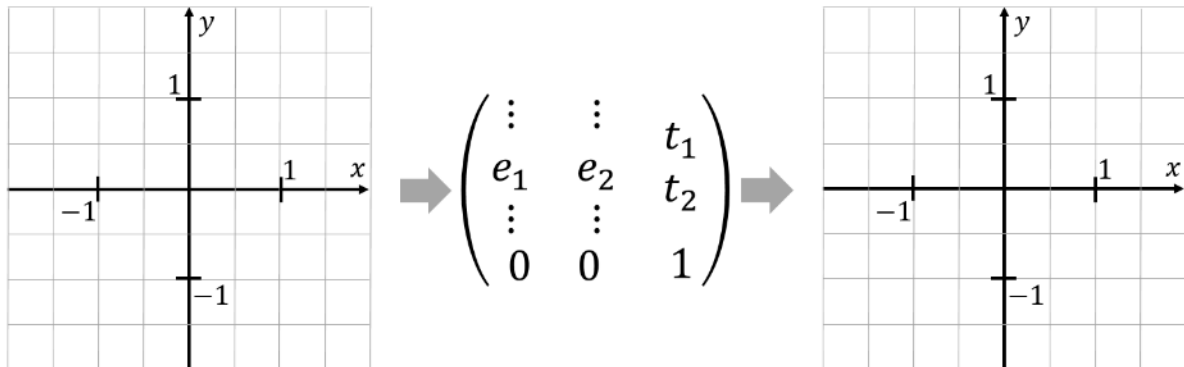
Erlangen, October 29th, 2020 _____
Signature

1	2	3	4	5	ALL
12	12	14	10	12	60

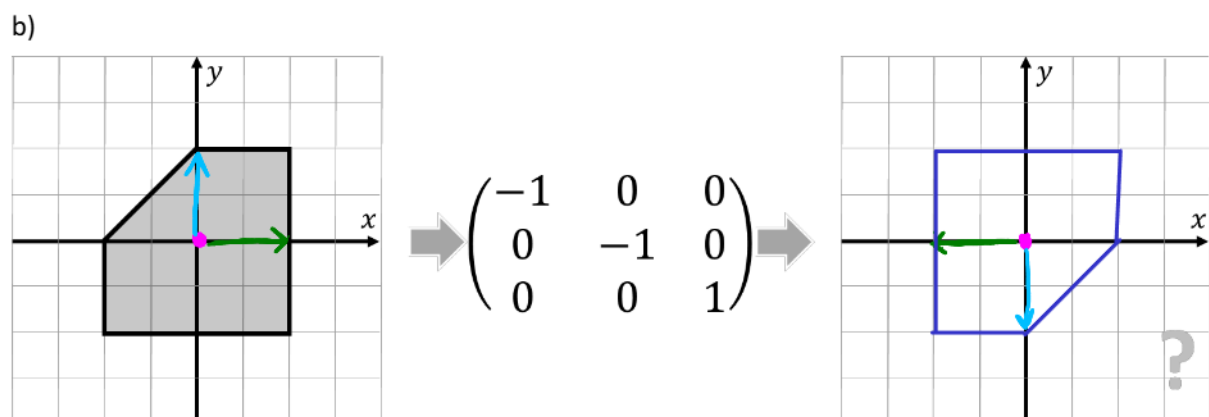
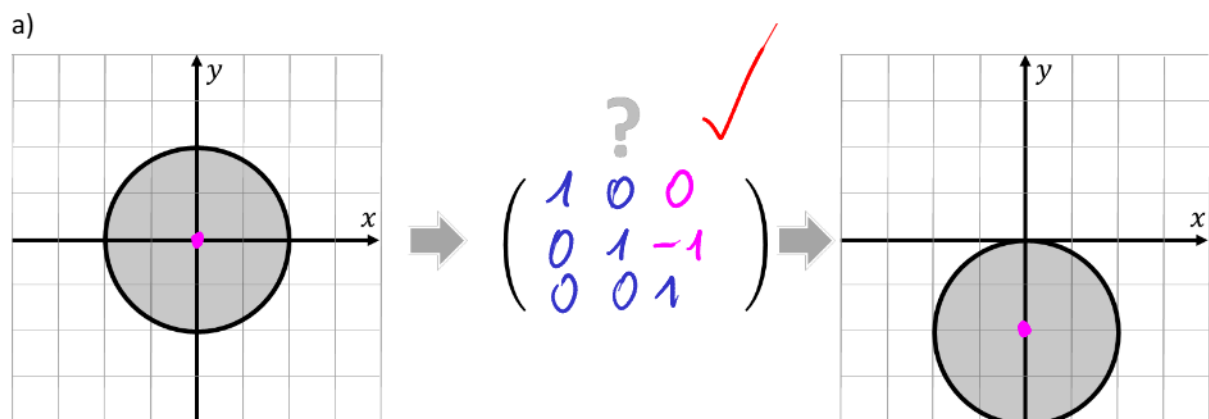
Grade:

Assignment 1: Affine Mappings in 2D [12 points]

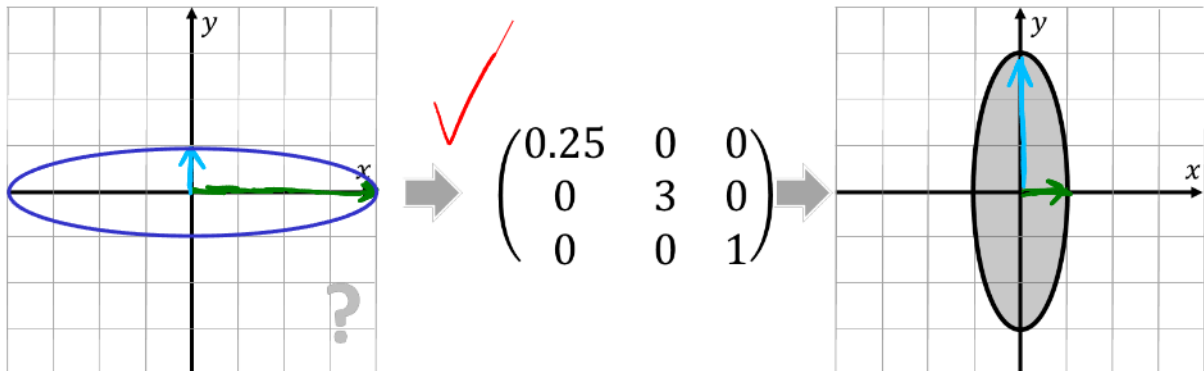
Affine mappings in 2D can be expressed using homogeneous coordinates and 3×3 matrices with the following structure, where $(\vec{e}_1, \vec{e}_2)$ is the linear part (image of base vectors) and (t_x, t_y) is the translation:



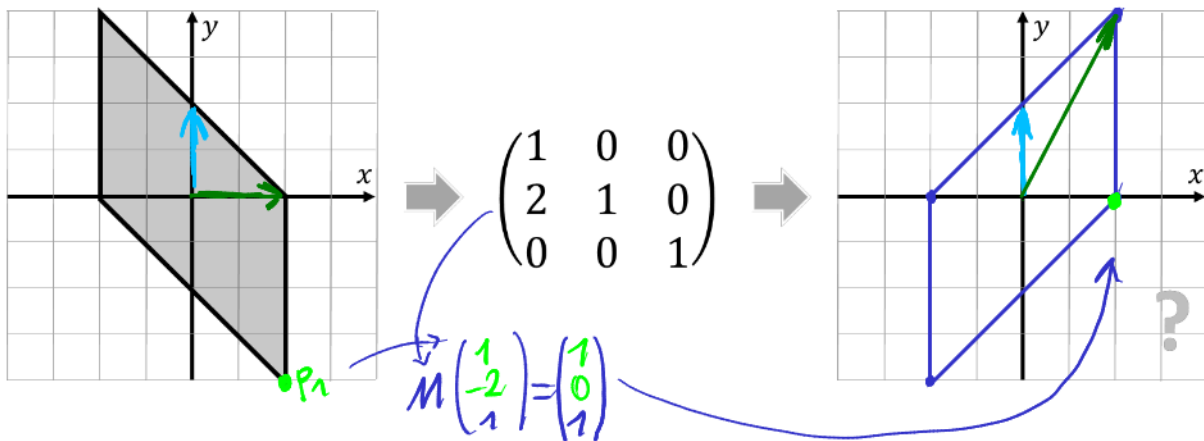
Each of the sub-assignments a) to e) deals with one mapping (in the coordinate system depicted above) and consists of two images (before and after mapping) and one matrix that has the desired effect. Complete each of the sub-assignments by filling in the missing part marked with a question mark.



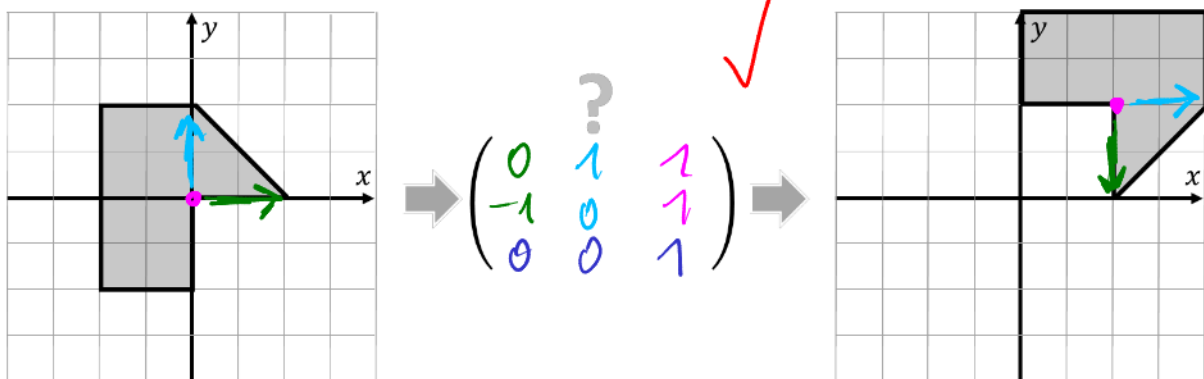
c)



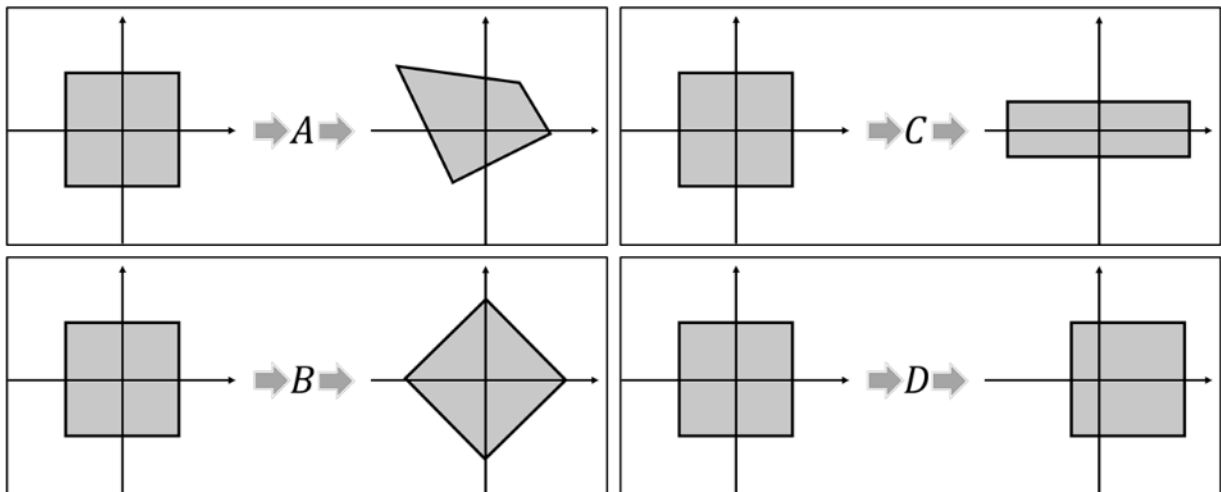
d)



e)



f) Below you see four mappings A, B, C and D, characterized by their result when applied to a square.



In the table below, each row describes one of the mappings above. For each mapping, state whether or not it is rigid, whether or not it is linear, and whether or not it is affine, by marking the respective properties with an X! Mappings can have more than one of the properties.

	rigid	linear	affine
<i>A</i>			
<i>B</i>	X	X	X
<i>C</i>		X	X
<i>D</i>	X		X



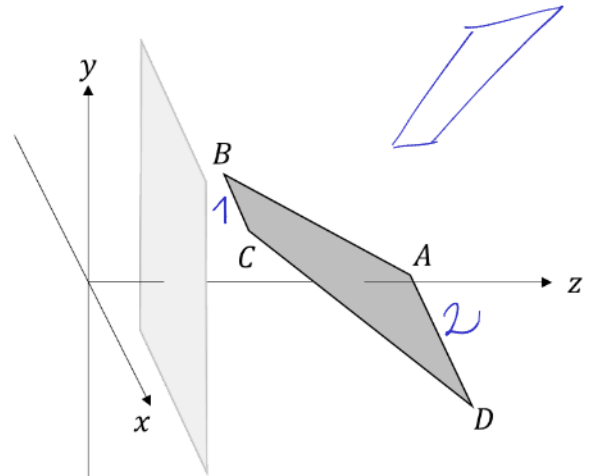
Assignment 2: Perspective Projection & Texture Mapping [12 points]

Given is a trapezoid $ABCD$ with the four vertices

$$A = (-1, -1, 4), B = (-0.5, 0.5, 2), \\ C = (0.5, 0.5, 2), D = (1, -1, 4).$$

We project the trapezoid perspectively (with the z -axis as view direction) using the following homogeneous matrix:

$$M = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



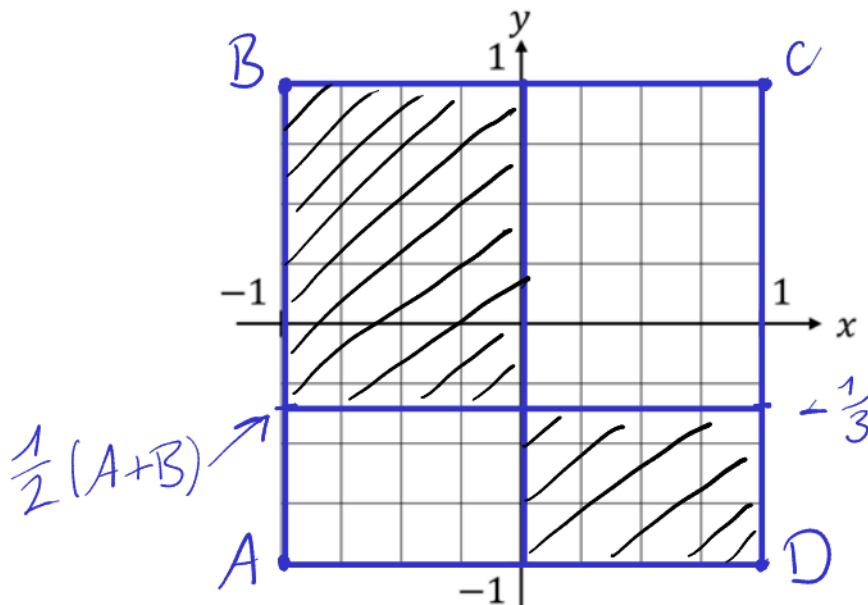
- a) Map the vertices to image space using matrix M and fill in the table with the mapping results. For all vertices, give both their two-dimensional image coordinates and their depth value after normalization!

before
normalization

Vertex	Image coordinates $\in [-1, 1]^2$	Depth after normalization $\in [-1, 1]$
A	$-1, -1$	$-3/4$
B	$-1, 1$	$-1/2$
C	$1, 1$	$-1/2$
D	$1, -1$	$-3/4$



- b) Sketch the resulting shape in image space into the following 8x8 pixel grid:



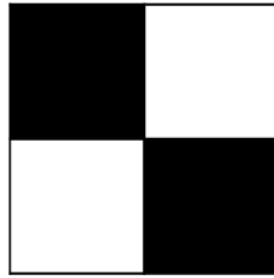
$$\text{Midpoint of AB: } \frac{1}{2}(A+B) = \begin{pmatrix} -0,75 \\ -0,25 \\ 3 \end{pmatrix}$$

$$\text{to image space: } \begin{pmatrix} -3 \\ -1 \\ -2 \\ 3 \end{pmatrix} \rightsquigarrow \begin{pmatrix} -1 \\ -1/3 \\ -2/3 \end{pmatrix} \quad \left. \begin{array}{l} \} (x) \\ \} (y) \\ \} z \end{array} \right\}$$

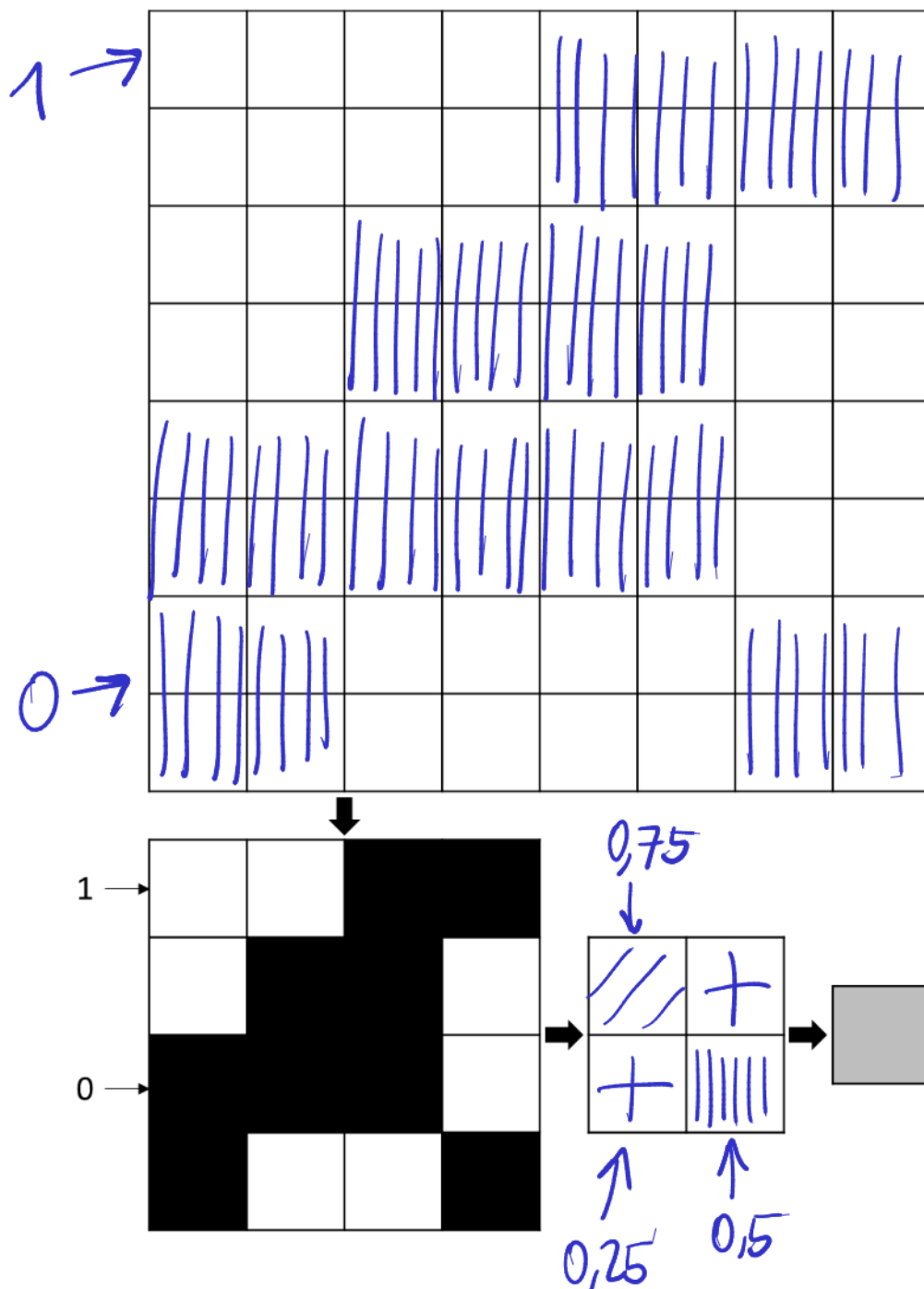
$$\text{Midpoint of CD: } \frac{1}{2}(C+D) = \begin{pmatrix} 0,75 \\ -0,25 \\ 3 \end{pmatrix}$$

$$\text{to image space: } \begin{pmatrix} 3 \\ -1 \\ -2 \\ 3 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 \\ -1/3 \\ -2/3 \end{pmatrix}$$

- c) Next, the following 2x2 texture should be mapped to the trapezoid in a perspective correct manner. Add a sketch of this mapped texture to your sketch in b).

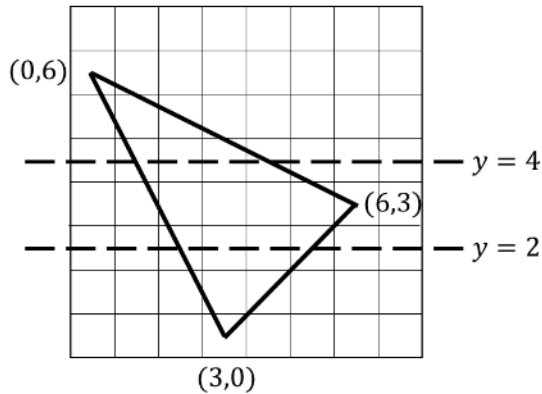


- d) Below you see a greyscale texture of a bird (roughly). Compute the missing levels of the MIP map pyramid this texture belongs to. Label your sketch with values between 0 (black) and 1 (white) as it is done for the given texture.



Assignment 3: Polygon Rasterization [14 points]

- a) The image below shows a triangle to be rasterized with the scanline algorithm. Fill in the values for the **Edge Table** on the right!



y_{lower}	x_{lower}	y_{upper}	$1/m$	next
0	3	6	$-\frac{1}{2}$	
0	3	3	1	
3	6	6	-2	

- b) Fill in the values for the **Active Edge Table** for scanlines $y = 2$ and $y = 4$!

active edge table for scanline $y = 2$:

x_{inter}	y_{upper}	$1/m$	next
2	6	$-\frac{1}{2}$	
5	3	1	

active edge table for scanline $y = 4$:

x_{inter}	y_{upper}	$1/m$	next
1	6	$-\frac{1}{2}$	
4	6	-2	

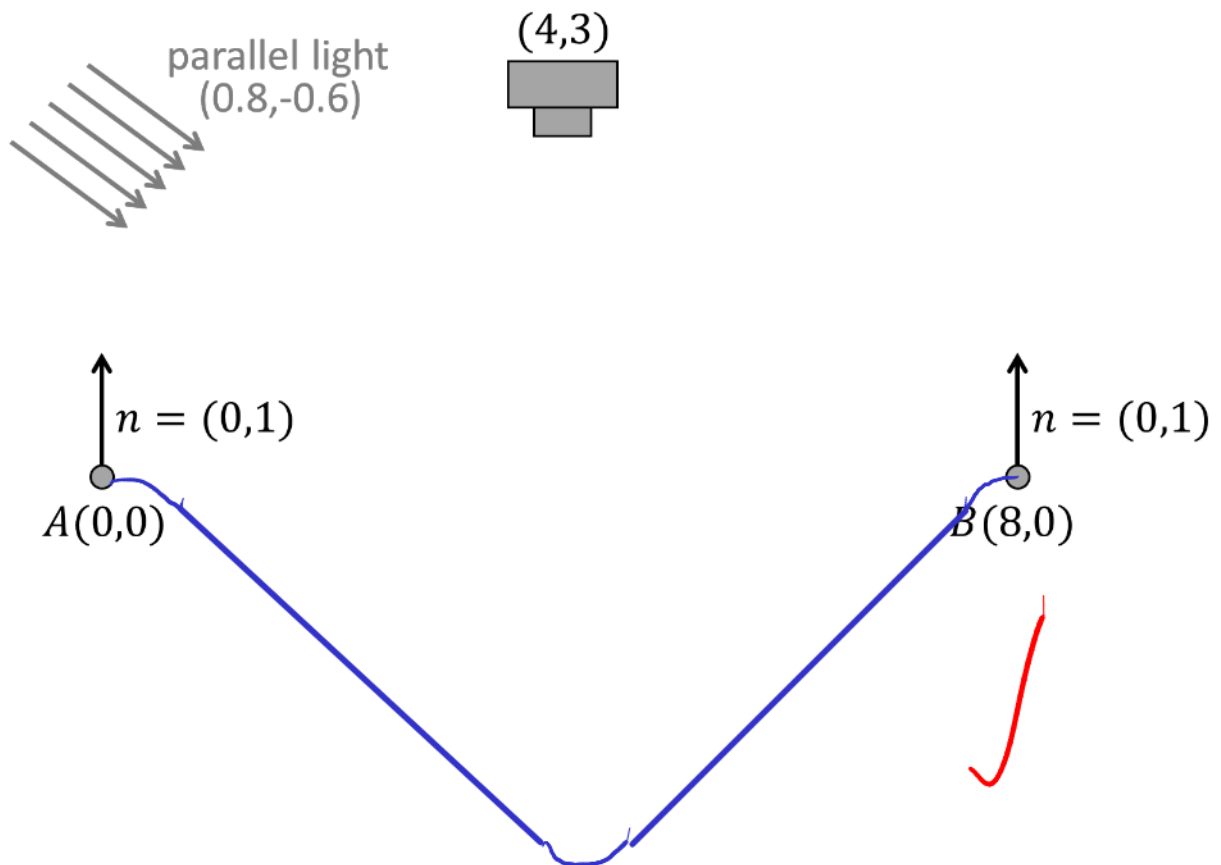
- c) Give a short explanation how Gouraud shading can be incorporated into the scanline algorithm!

- calculate color of vertices
 - linear interpolation along edges
 - during scanline algorithm:
 when drawing a line between two hitpoints (points of 2 edges)
 use linear interpolation of the hitpoints colors

[see also: exam 2017 WS]

Assignment 4: Phong Lighting & Shading [10 points]

The image below shows two points A and B belonging to a common surface. The surface is lit by a parallel light source with rays in direction $(0.8, -0.6)$ and captured by a camera located at $(4, 3)$. Both points have a normal $n = (0, 1)$.



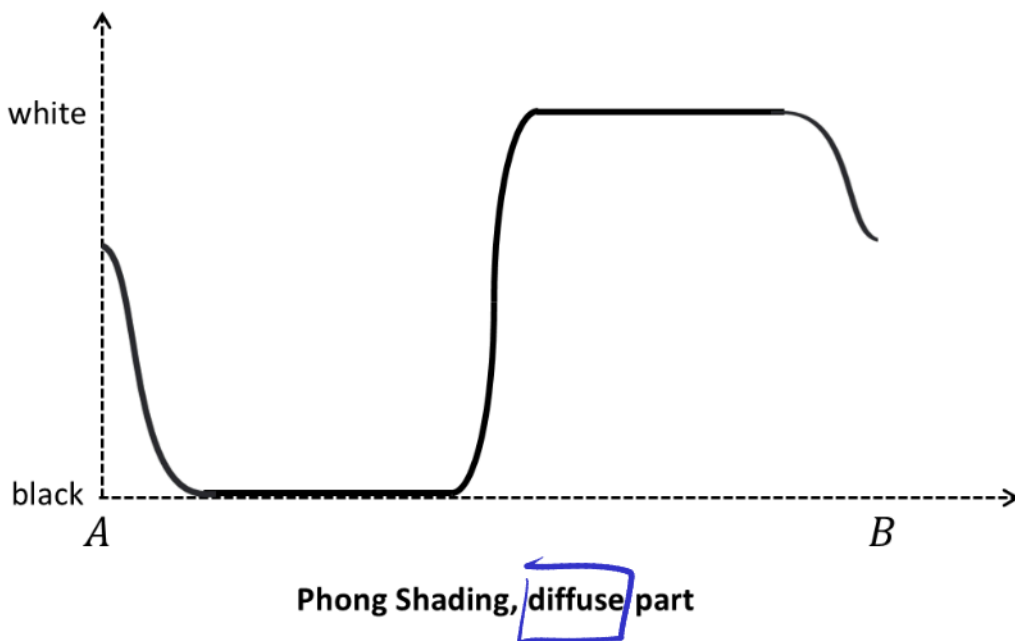
- a) For the two points A and B , write down the respective light vector l , view vector v and reflection vector r !

	A	B
l	$(-0,8 \mid 0,6)$	$(-0,8 \mid 0,6)$
v	$(0,8 \mid 0,6)$	$(-0,8 \mid 0,6)$
r	$(0,8 \mid 0,6)$	$(0,8 \mid 0,6)$

- b) Let's assume the following Phong Lighting parameters for the surface:
 $k_{amb} = (0,0,0)$, $k_{diff} = (1,0,0)$, $k_{spec} = (0,1,0)$, $n_{shiny} = 2$
 The parallel light is of intensity $I_{in} = (1,1,1)$.
 Compute the colors of the two points A and B!

$$\begin{aligned}
 L &= L_{amb} + L_{diff} + L_{spec}, \quad nol = 0,6 \\
 &= k_{amb} I_{amb} + k_{diff} I_{in} (nol) + k_{spec} I_{in} (vor)^{n_{shiny}} \\
 L_A &= 0 + \begin{pmatrix} 0,6 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0,6 \\ 1 \\ 0 \end{pmatrix} \\
 L_B &= 0 + \begin{pmatrix} 0,6 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0,28^2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0,6 \\ 0,28^2 \\ 0 \end{pmatrix} \\
 r \cdot v_A &= 1, \quad r \cdot v_B = -0,28
 \end{aligned}$$

- c) Next, assume that the surface between A and B is made up of purely **diffuse white** material ($k_{amb} = k_{spec} = (0,0,0)$, $k_{diff} = (1,1,1)$). Below, you see a sketch of the lighting over the surface when shaded with Phong Shading.
 In the figure above, sketch the surface between A and B! Take into account the information about the normals (from the figure) and the shape of the curve for the diffuse lighting!



Assignment 5: Color Models [12 points]

- a) The table below lists gives some colors, each either by its name or by its representation in one of the three color models **RGB**, **CMYK** and **HSV**. Fill in the empty cells so that each row gives all representations for one color. Take care that for CMYK the **K** value is as **large as possible**. In HSV it is possible that some values are not uniquely defined, in this case you can choose an arbitrary value.

Hint: In HSV, the angles for red, green and blue are 0° , 120° and 240° , respectively.

Color Name	RGB	CMYK	HSV
black	(0, 0, 0)	(0, 0, 0, 1)	(0° , 0, 0)
white	(1, 1, 1)	(0, 0, 0, 0)	(π° , 0, 1)
red	(1, 0, 0)	(0, 1, 1, 0)	(0° , 1, 1)
bright red	(1, 0.5, 0.5)	(0, 0.5, 0.5, 0)	(0° , 0.5, 1)
yellow	(1, 1, 0)	(0, 0, 1, 0)	(60° , 1, 1)
grey	(0.5, 0.5, 0.5)	(0, 0, 0, 0.5)	(π° , 0, 0.5)
dark green	(0, 0.5, 0)	(0.5, 0, 0.5, 0.5)	(120° , 1, 0.5)
bright cyan	(0.5, 1, 1)	(0.5, 0, 0, 0)	(180° , 0.5, 1)

- b) Which triplet means the same color both when evaluated in the RGB color model and when evaluated in the CMY color model?

(0.5 / 0.5 / 0.5)

- c) In HSV, some colors are not uniquely defined and can be encoded with different triplets. Give an example by providing two HSV triplets which encode the same color!

